COMP5331 (16 Oct)

Show that if X is conditionally independent of Y given Z, $P(X, Y|Z) = P(X|Z) \cdot P(Y|Z)$

Proof: $P(X, Y|Z) = \frac{P(X, Y, Z)}{P(Z)}$ (by ①)

$= \frac{P(X, Y, Z)}{P(Y, Z)} \cdot \frac{P(Y, Z)}{P(Z)}$

$= P(X|Y, Z) \cdot P(Y|Z)$

~~(by ②)~~ and (by ①)

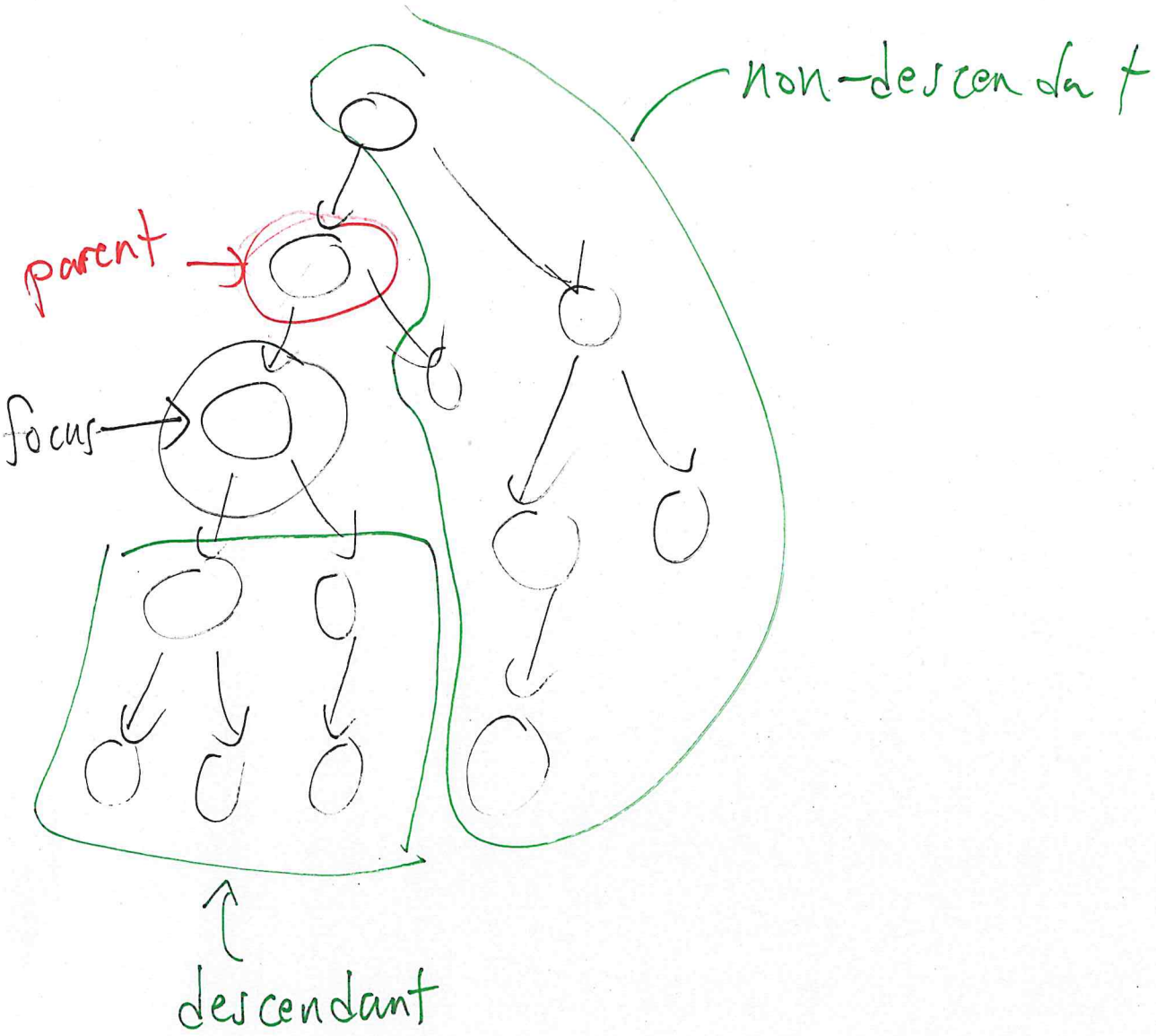
$= P(X|Z) \cdot P(Y|Z)$

(by ②)

① $P(A|B) = \frac{P(A, B)}{P(B)}$

② $P(X|Y, Z) = P(X|Z)$

(if X is conditionally independent of Y given Z


non-descendant
parent
focus
descendant

Goal: $P(HD = Yes)$

E=Yes, D=Healthy

$= P(HD = Yes \mid Case\ 1)\, P(Case\ 1)$

E=Yes, D=Unhealthy

$+ P(HD = Yes \mid Case\ 2)\, P(Case\ 2)$

E=No, D=Healthy

$+ P(HD = Yes \mid Case\ 3)\, P(Case\ 3)$

E=No, D=Unhealthy

$+ P(HD = Yes \mid Case\ 4)\, P(Case\ 4)$

$= 0.25 \times P(E = Yes,\ D = Healthy)$

$+ 0.45 \times P(E = Yes,\ D = Unhealthy)$

$+ 0.55 \times P(E = No,\ D = Healthy)$

$+ 0.75 \times P(E = No,\ D = Unhealthy)$

0.7

$= 0.25 \times P(E = Yes) \times P(D = Healthy)$

0.25

$+ 0.45 \times P(E = Yes) \times P(D = Un_)$

$+ 0.55 \times P(E = No) \times P(D = He_)$

0.75

$+ 0.75 \times P(E = No) \times P(D = Unh_)$

0.3

$= 0.49$

$P(HD = No)$

$= 1 - P(HD = Yes)$

$= 1 - 0.49$

$= 0.51$

Concl: It is very likely that this patient will not suffer from Heat Diseases

$$P(HD = Yes)$$

$$= P(HD = Yes \mid Gender = M)\, P(Gender = M) + P(HD = Yes \mid Gender = F)\, P(Gender = F)$$

$$P(HD = Yes)$$

$$= P(HD = Yes \mid Case\ 1)\, P(Case\ 1) + P(HD = Yes \mid Case\ 2)\, P(Case\ 2)$$

$$P(HD = Yes \mid E, F)$$

$$= P(HD = Yes \mid Case\ 1, E, F)\, P(Case\ 1 \mid E, F) + P(HD = Yes \mid Ca\ 2, E, F)\, P(Ca\ 2 \mid E, F)$$

eg. 1 $P(HD = Yes) = 0.49$

$P(HD = No) = 0.51$

Goal: $P(HD = Yes \mid BP = High)$

$$= \frac{P(BP = High \mid HD = Yes)\, P(HD = Yes)}{P(BP = High)}$$

$$= \frac{0.85 \times 0.49}{P(BP = High)}$$

$$= \frac{0.85 \times 0.49}{\boxed{\quad}}$$

$$= \boxed{\quad} = 0.8033$$

$$P(HD = No \mid BP = High)$$

$$= 1 - 0.8033$$

$$= 0.1967$$

$P(BP = High)$ — HD=Yes, HD=No

$= P(BP = High \mid (Case\ 1)) \times P((Case\ 1))$

$+ P(BP = High \mid (Case\ 2)) \times P((Case\ 2))$

$= 0.85 \times 0.49$

$+ 0.2 \times 0.51$

$= \boxed{\quad}$

Concl: It is very likely that this patient will suffer from Heart Disease

Goal

BP

↓

HD

GNew

HD

↓

BP

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

$$P(A|B, E, F) = \frac{P(B|A, E, F) \times P(A|E, F)}{P(B|E, F)}$$

Goal: $P(HD = Yes \mid E = Yes, D = Healthy, BP = High)$

$= P(HD = Yes \mid BP = High, E = Yes, D = Healthy)$

$= \dfrac{P(BP = High \mid HD = Yes, E = Yes, D = Healthy) \times P(HD = Yes \mid E = Yes, D = Healthy)}{P(BP = High \mid E = Yes, D = Healthy)}$

$= \dfrac{P(BP = High \mid HD = Yes) \times 0.25}{P(BP = High \mid E = Yes, D = Healthy)}$

$= \dfrac{0.85 \times 0.25}{P(BP = High \mid E = Yes, D = Healthy)}$

$= \dfrac{0.85 \times 0.25}{0.3625}$

$= 0.5862$

$P(HD = No \mid E = Yes, D = Healthy, BP = High)$

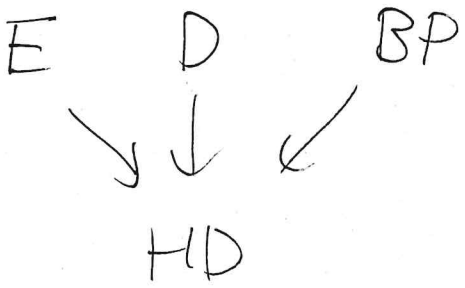
$= 1 - P(HD = Yes \mid E = Yes, D = Healthy, BP = High)$

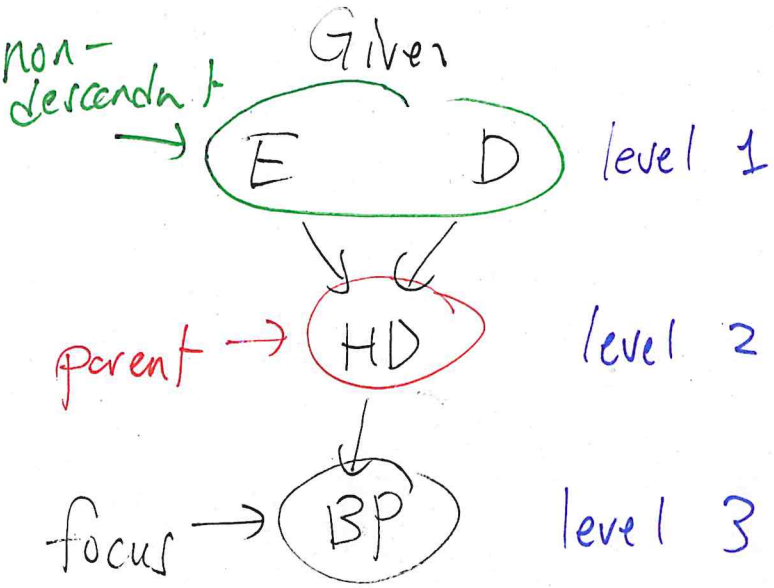
$= 1 - 0.5862$

$= 0.4138$

Concl: It is likely that this patient will suffer from Heart Disease

Goal


E
D
BP
HD


non-descendant
Given
E
D
level 1
parent
HD
level 2
focus
BP
level 3

$P(BP=High \mid E=Yes, D=Heathy)$

$= P(BP=High \mid E=Yes, D=Healthy)$

$= P(BP=High \mid HD=Yes, E=Yes, D=Healthy) \times$

$P(HD=Yes \mid E=Yes, D=Healthy)$

$+ P(BP=High \mid HD=No, E=Yes, D=Healthy) \times$

$P(HD=No \mid E=Yes, D=Healthy)$

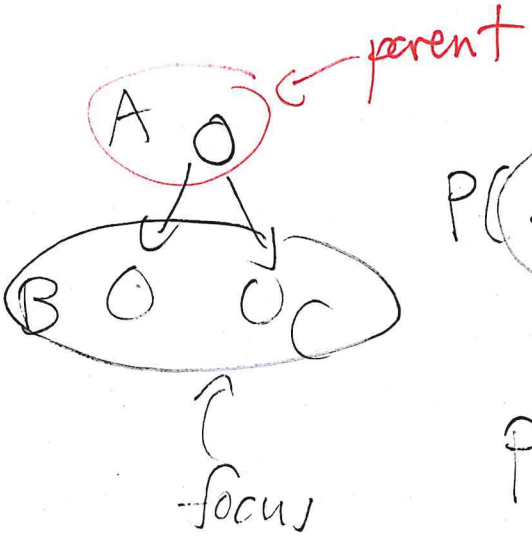
$= P(BP=High \mid HD=Yes) \times \underline{0.25}$

$+ P(BP=High \mid HD=No) \times \underline{0.75}$

$= \underline{0.85} \times 0.25$

$+ \underline{0.2} \times 0.75$

$= \boxed{0.7}$



$$P(B, C \mid A) = P(B \mid A) \times P(C \mid A)$$

~~$$P(B, C) = P(B) \times P(C)$$~~